

Radiative Transitions of Strange Heavy Axial Mesons from Light-Cone QCD Sum Rules

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Abstract

We present a light-cone QCD sum-rule analysis of $D_{s1}(2460) \rightarrow D_s \gamma$, $D_{s1}(2536) \rightarrow D_s \gamma$, and $B_{s1}(5830) \rightarrow B_s \gamma$. The calculation keeps explicit strange-quark-mass terms and includes hard photon emission from both quark lines, two-particle photon distribution amplitudes, and the three-particle photon-amplitude corrections induced by the background-gluon term in the quark propagator. We separate the basis-current sum rules from the physical-current residue normalization. For the charm-strange channel the physical residues are projected from an $AA/AB/BB$ two-point matrix at the previously determined mixing angle; the bottom-strange residues are obtained from the direct heavy-flavour continuation of the same matrix calculation. Using the modern lattice normalization of the leading-twist photon distribution amplitude, we find $\Gamma[D_{s1}(2460) \rightarrow D_s \gamma] = 34.96[28.06, 43.53]$ keV, $\Gamma[D_{s1}(2536) \rightarrow D_s \gamma] = 0.988[0.092, 3.555]$ keV, and $\Gamma[B_{s1}(5830) \rightarrow B_s \gamma] = 8.702[6.474, 11.23]$ keV. The operator-weighted gluonic and electromagnetic convolutions are evaluated with the exact off-shell double-Borel numerator reduction and agree term-by-term between the Python and Mathematica implementations.

1 Introduction

Radiative transitions of positive-parity strange heavy mesons probe the interplay of heavy-quark spin symmetry, light-quark dynamics, and threshold effects. The two charm-strange axial states, $D_{s1}(2460)$ and $D_{s1}(2536)$, are especially useful because the lower state has a measured radiative branching fraction [1], while the upper state has a measured total width but no prominent $D_s \gamma$ signal [2]. The bottom-strange state $B_{s1}(5830)$ is observed in hadronic spectroscopy [3]; its radiative decay has not yet been measured.

The aim of this paper is not only to quote updated numbers, but also to make the normalization of the calculation unambiguous. The light-cone OPE gives transition form factors for basis currents. The conversion to the physical 1^+ states requires the residues of the physical mixed currents. We therefore organize the analysis in stages: first compute the axial and tensor basis amplitudes, then add the photon-DA corrections, and finally apply the physical-current normalization.

The hierarchy between $D_{s1}(2460) \rightarrow D_s \gamma$ and $D_{s1}(2536) \rightarrow D_s \gamma$ was recently emphasized by Bondar and by Bondar and Milstein [4, 5]. Their quark-model explanation is that the physical $J = 1$ states are mixed spin configurations and that the E1 amplitudes enter with different relative signs. Below we show that the same mechanism appears directly in the LCSR decomposition of the physical amplitudes.

Two photon-input scenarios are considered throughout. The first is the legacy magnetic-susceptibility normalization of the photon distribution amplitudes [6, 7]. The second uses the recent lattice normalization of the leading-twist photon distribution amplitude [8]. The latter is our preferred modern scenario; the former is kept to show continuity with older light-cone sum-rule analyses.

2 Hadronic Conventions

We use $p' = p + q$, where p' is the momentum of the initial axial meson, p is the momentum of the final pseudoscalar meson, and q is the real-photon momentum. The pseudoscalar interpolating current is

$$J_5 = \bar{s} i \gamma_5 Q, \quad Q = c, b. \quad (1)$$

For the axial channel we introduce two basis currents,

$$J_\mu^A = \bar{s} \gamma_\mu \gamma_5 Q, \quad (2)$$

$$J_\mu^B = \frac{i}{m_Q + m_s} \bar{s} \sigma_{\mu\alpha} p'^\alpha \gamma_5 Q. \quad (3)$$

The momentum in the tensor current is the initial-state momentum p'^α . The physical-current combinations are

$$J_{1\mu} = \sin \theta J_\mu^A + \cos \theta J_\mu^B, \quad (4)$$

$$J_{2\mu} = \cos \theta J_\mu^A - \sin \theta J_\mu^B, \quad (5)$$

We use the sector-specific results of our dedicated mixing-angle study [9],

$$\theta_{D_s} = 26.6 \pm 0.6^\circ, \quad \theta_{B_s} = 38.5 \pm 0.1^\circ. \quad (6)$$

The heavy-quark-limit value 35.3° and the interval $25^\circ \leq \theta \leq 45^\circ$ are retained only as benchmark and sensitivity tests; the latter is not interpreted as a statistical confidence interval.

The radiative matrix element is parametrized as

$$\langle P(p) \gamma(q, \varepsilon) | A(p', \eta) \rangle = eG [(\eta \cdot \varepsilon^*)(p \cdot q) - (\eta \cdot q)(p \cdot \varepsilon^*)]. \quad (7)$$

The corresponding width is

$$\Gamma(A \rightarrow P \gamma) = \frac{\alpha_{\text{em}}}{3} G^2 \left(\frac{m_A^2 - m_P^2}{2m_A} \right)^3. \quad (8)$$

3 Light-Cone Sum Rule

The starting correlation function is

$$\Pi_\mu(p, q) = i \int d^4x e^{ip \cdot x} \langle \gamma(q, \varepsilon) | T \{ \bar{s}(x) \Gamma_\mu Q(x) \bar{Q}(0) i \gamma_5 s(0) \} | 0 \rangle, \quad (9)$$

where Γ_μ is either J_μ^A or J_μ^B . We project onto the gauge-invariant E1 tensor

$$T_{\mu\nu} = (p \cdot q) g_{\mu\nu} - q_\mu p_\nu. \quad (10)$$

The OPE is constructed with quark propagators in an external electromagnetic field. Both hard photon emission and soft photon distribution-amplitude terms are retained. In addition, the background-gluon insertion in the propagator generates the three-particle photon-DA terms. The ordinary vacuum-condensate pieces of the propagator are not inserted as separate three-point corrections: in the photon LCSR they enter through photon matrix elements and normalizations such as $\chi_s \langle \bar{s}s \rangle$. This is separate from the local two-point sum rules used to determine decay constants, where ordinary SVZ condensates are part of the OPE.

After Borel transformation and continuum subtraction the basis-current amplitudes are denoted G_A and G_B . The charm physical couplings are then formed as

$$G_{2460} = \frac{\sin \theta f_A G_A + \cos \theta f_B G_B}{f_1}, \quad (11)$$

$$G_{2536} = \frac{\cos \theta f_A G_A - \sin \theta f_B G_B}{f_2}. \quad (12)$$

Here f_1 and f_2 are obtained by projecting the normalized-current $AA/AB/BB$ two-point QCD sum-rule matrix at the external D_s angle. The LO+ $d = 3+d = 5$ Monte Carlo gives

$$f_1 = 0.4046[0.3799, 0.4291] \text{ GeV}, \quad f_2 = 0.1677[0.1594, 0.1756] \text{ GeV}. \quad (13)$$

Diagonalizing this truncated matrix gives $30.63[29.94, 31.29]^\circ$ only as a diagnostic. The normalized off-diagonal residual after the external-angle projection is $0.183[0.142, 0.225]$.

Wang’s current arXiv v4 result is $f_A(D_{s1}) = 0.345 \pm 0.017 \text{ GeV}$ [10]; the former $0.245 \pm 0.017 \text{ GeV}$ entry was explicitly corrected. Reproducing that number is neither necessary nor sufficient to verify the mixed residues: Wang uses a pure J_A current and a one-pole ansatz, whereas our AA channel contains the two physical poles after the mixed-current projection. We therefore make the comparison at two levels. The direct LO+ $d = 3+d = 5$ AA OPE evaluated in Wang’s window gives $0.2686[0.2607, 0.2768] \text{ GeV}$. Independently, projecting our two physical poles back onto J_A gives a Borel-weighted one-pole-equivalent residue $0.2303[0.2194, 0.2418] \text{ GeV}$. The analytic projection is written in Appendix B. Both numbers are normalization diagnostics; neither is used as an input or presented as a reproduction of Wang’s complete OPE.

For the bottom-strange channel we evaluate the same normalized-current $AA/AB/BB$ OPE with $m_Q = m_b$ and project it at the previous-study angle $\theta_{B_s} = 38.5 \pm 0.1^\circ$. In the selected two-point window $M^2 = 6.50\text{--}7.30 \text{ GeV}^2$, separate thresholds are fitted to the model lower pole and the observed $B_{s1}(5830)$. The direct Monte Carlo gives

$$f_1^{(B_s)} = 0.5358[0.5096, 0.5586] \text{ GeV}, \quad f_2^{(B_s)} = 0.08896[0.08496, 0.09278] \text{ GeV}. \quad (14)$$

Of 2000 generated samples, 1757 satisfy all stated stability criteria. No Pullin–Zwicky decay constant and no off-diagonal overlap closure enters the final bottom normalization; their earlier use is retained only in the internal historical comparison.

4 Numerical Inputs and Scale Convention

Table 1 lists the main inputs. Hadron masses, quark masses, charges, and α_{em} are taken from the PDG review and online listings [11, 12]. The pseudoscalar decay constants are taken in the lattice context summarized by the published FLAG review [13]. Photon DAs follow Ball, Braun and Kivel [6]; the legacy susceptibility comparison follows Rohrwild [7]; the modern photon normalization follows the lattice result of Bacchio et al. [8]. The input bookkeeping is kept in `inputs_table.csv` and in the generated citation map `outputs/input_citation_map.csv`.

The photon-DA normalizations are quoted at $\mu = 1 \text{ GeV}$, while the quark masses are the usual $\overline{\text{MS}}$ inputs, e.g. $m_s(2 \text{ GeV})$ and $m_c(m_c)$. This is a standard phenomenological convention rather than a fully matched single-scale setup. We therefore treat the lattice photon-normalized scenario as the preferred modern input and retain the susceptibility scenario as a comparison with older analyses.

5 Results

The final numerical results are summarized in Table 3. We quote medians and 16–84 percentile intervals rather than Gaussian means and standard deviations. This is important because

Quantity	Symbol	Value	Source/use
quark masses	m_c, m_b, m_s	1.27(2), 4.18(3), 0.093(11) GeV	PDG; explicit m_s terms kept
charm masses	$M_{2460}, M_{2536}, m_{D_s}$	2.4595, 2.53511, 1.96835 GeV	PDG
bottom masses	M_{5830}, m_{B_s}	5.82870, 5.36692 GeV	PDG/CMS
lower B_{s1} diagnostic mass	$M_{B_{s1}}^{\text{low}}$	5.750(26) GeV	lattice-inspired diagnostic [14]
pseudoscalar decay constants	f_{D_s}, f_{B_s}	0.2499, 0.2303 GeV	FLAG/lattice context
charm mixed residues	f_1, f_2	0.4046[0.3799, 0.4291], 0.1677[0.1594, 0.1756] GeV	fixed-angle $AA/AB/BB$ projection
pure axial benchmark	$f_A(D_{s1})$	0.345(17) GeV	corrected Wang v4 pure-current check
bottom physical constants	$f_1^{(B_s)}, f_2^{(B_s)}$	0.5358[0.5096, 0.5586], 0.08896[0.08496, 0.09278] GeV	direct $AA/AB/BB$ projection
bottom two-point window	$M_{2\text{pt}}^2$	6.50–7.30 GeV ²	pole/OPE/mass stability
legacy photon input	$\chi_s(1 \text{ GeV})$	+3.15(30) GeV ⁻²	comparison scenario
lattice photon input	$f_{\gamma,s}^\perp(1 \text{ GeV})$	−55.1(1.9) MeV	preferred scenario

Table 1: Main numerical inputs used in the analysis.

Quantity	Current	Central value	Interval	Interpretation
Wang v4 f_A	pure J_A , one pole	0.345	± 0.017	external benchmark
our direct AA OPE	pure J_A	0.2686	[0.2607, 0.2768]	truncation benchmark
projected AA equivalent	J_A , two poles	0.2303	[0.2194, 0.2418]	one-pole-equivalent diagnostic
f_1	mixed J_1	0.4046	[0.3799, 0.4291]	not directly Wang-comparable
f_2	mixed J_2	0.1677	[0.1594, 0.1756]	not directly Wang-comparable
$\sin \theta f_1$	$J_A \rightarrow D_{s1}(2460)$	0.1812	[0.1692, 0.1930]	projection diagnostic
$\cos \theta f_2$	$J_A \rightarrow D_{s1}(2536)$	0.1498	[0.1424, 0.1573]	projection diagnostic

Table 2: Pure-current and projected-current comparison. Wang’s number, our direct truncated AA OPE, and the projected two-pole equivalent do not share the same OPE truncation and pole model. The table is therefore a normalization benchmark, not a direct validation of either mixed residue. Decay constants are in GeV.

the $D_{s1}(2536)$ and $B_{s1}(5830)$ amplitudes can pass close to zero; their width distributions are therefore non-Gaussian.

Sector	Decay	Photon input	Mixing	Normalization	G (GeV ⁻¹)	Γ (keV)
D_s	$D_{s1}(2460) \rightarrow D_s \gamma$	legacy χ_s	26.6(6)°	$f_1 = 0.405[0.381, 0.429]$	−0.542[−0.611, −0.479]	61.66[48.37, 78.46]
D_s	$D_{s1}(2536) \rightarrow D_s \gamma$	legacy χ_s	26.6(6)°	$f_2 = 0.168[0.159, 0.175]$	−0.0790[−0.153, −0.00266]	2.220[0.226, 7.379]
D_s	$D_{s1}(2460) \rightarrow D_s \gamma$	lattice $f_{\gamma,s}^\perp$	26.6(6)°	$f_1 = 0.405[0.379, 0.430]$	−0.408[−0.455, −0.365]	34.96[28.06, 43.53]
D_s	$D_{s1}(2536) \rightarrow D_s \gamma$	lattice $f_{\gamma,s}^\perp$	26.6(6)°	$f_2 = 0.168[0.160, 0.175]$	+0.0518[+0.00244, +0.1067]	0.988[0.092, 3.555]
B_s	$B_{s1}(5830) \rightarrow B_s \gamma$	legacy χ_s	38.5(1)°	$f_2^{(B_s)} = 0.0894[0.0854, 0.0929]$	−0.269[−0.302, −0.235]	15.37[11.75, 19.31]
B_s	$B_{s1}(5830) \rightarrow B_s \gamma$	lattice $f_{\gamma,s}^\perp$	38.5(1)°	$f_2^{(B_s)} = 0.0888[0.0845, 0.0926]$	−0.203[−0.230, −0.175]	8.702[6.474, 11.23]
B_s	$B_{s1}^{\text{low}}(5750) \rightarrow B_s \gamma$	legacy χ_s	38.5(1)°	$f_1^{(B_s)} = 0.533[0.508, 0.557]$	+0.216[+0.181, +0.249]	5.435[3.949, 7.588]
B_s	$B_{s1}^{\text{low}}(5750) \rightarrow B_s \gamma$	lattice $f_{\gamma,s}^\perp$	38.5(1)°	$f_1^{(B_s)} = 0.536[0.510, 0.558]$	+0.308[+0.288, +0.332]	11.34[9.144, 14.06]

Table 3: Final form-factor and width predictions. The lattice photon input is the preferred modern normalization; the legacy susceptibility input is kept as a comparison with older photon-DA analyses. The $B_{s1}^{\text{low}}(5750)$ entries are lower-partner diagnostics, not observed resonances. Both sectors use direct $AA/AB/BB$ two-point residues.

The preferred lattice-normalized result for $D_{s1}(2460)$ is a tens-of-keV partial width. In contrast, $D_{s1}(2536)$ is highly cancellation sensitive: the same axial and tensor basis amplitudes enter with the opposite physical combination in Eq. (12), driving the amplitude close to zero for a wide part of the accepted input range. The observed $B_{s1}(5830)$ behaves similarly in the bottom sector. Its median width is at the keV scale after the direct physical-current normalization.

Table 4 makes this cancellation explicit. The entries are the two normalized pieces of Eq. (12). For $D_{s1}(2460)$ both pieces have the same sign and add constructively. For $D_{s1}(2536)$ they have opposite signs and nearly cancel. This is the LCSR version of the Bondar–Milstein explanation of why the $D_{s1}(2536)$ radiative mode is difficult to observe.

The suppression is not a phase-space effect or an exact selection rule. At the nominal angle

Case	Mixing	A_{2460}	B_{2460}	G_{2460}	A_{2536}	B_{2536}	G_{2536}	$ G_{2460}/G_{2536} $
previous-study nominal	26.6°	-0.0749	-0.3365	-0.4113	-0.3648	+0.4110	+0.0463	8.89
truncated-matrix diagnostic	30.61°	-0.0850	-0.3234	-0.4084	-0.3597	+0.4789	+0.1192	3.43
HQET benchmark	35.3°	-0.0966	-0.3070	-0.4035	-0.3358	+0.5350	+0.1992	2.03

Table 4: Central lattice-photon benchmark at the midpoint of the selected transition window. The 2460 pieces remain constructive and the 2536 pieces remain destructive; the angle changes the distance from the cancellation near 23.96° . All G values are in GeV^{-1} .

the two high-state pieces are individually of order 0.4 GeV^{-1} ,

$$G_{2536}^{(A)} = -0.3648 \text{ GeV}^{-1}, \quad G_{2536}^{(B)} = +0.4110 \text{ GeV}^{-1}, \quad (15)$$

but their sum is only $+0.0463 \text{ GeV}^{-1}$. Since the width is proportional to $|G|^2 q_\gamma^3$, this amplitude-level compensation becomes a much stronger width suppression. It is also why calculations with a different mixing angle, current normalization, or relative axial-tensor basis amplitude can obtain substantially larger widths: at the HQET benchmark angle the same central OPE gives $G_{2536} = 0.199 \text{ GeV}^{-1}$ and $\Gamma_{2536} = 12.3 \text{ keV}$. The robust prediction is therefore the destructive-interference mechanism and the hierarchy relative to the lower state; the precise sub-keV median is intrinsically cancellation sensitive.

The lower B_{s1} row in Table 3 is retained only as a model-state diagnostic. It is not an experimentally established resonance and is not included in the final experiment-theory comparison.

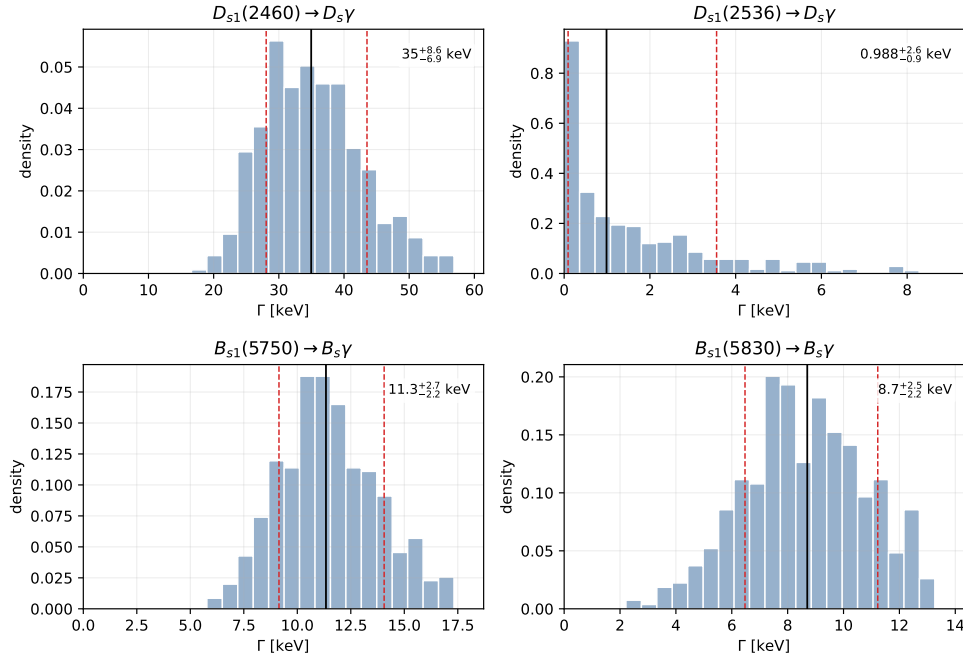


Figure 1: Final-window Monte Carlo width distributions with the preferred lattice photon normalization and the sector-specific previous-study mixing angles.

6 Reproducibility

Table 5 records the source of each headline table and figure. The machine-readable version is `outputs/manuscript_reproducibility_map.csv`.

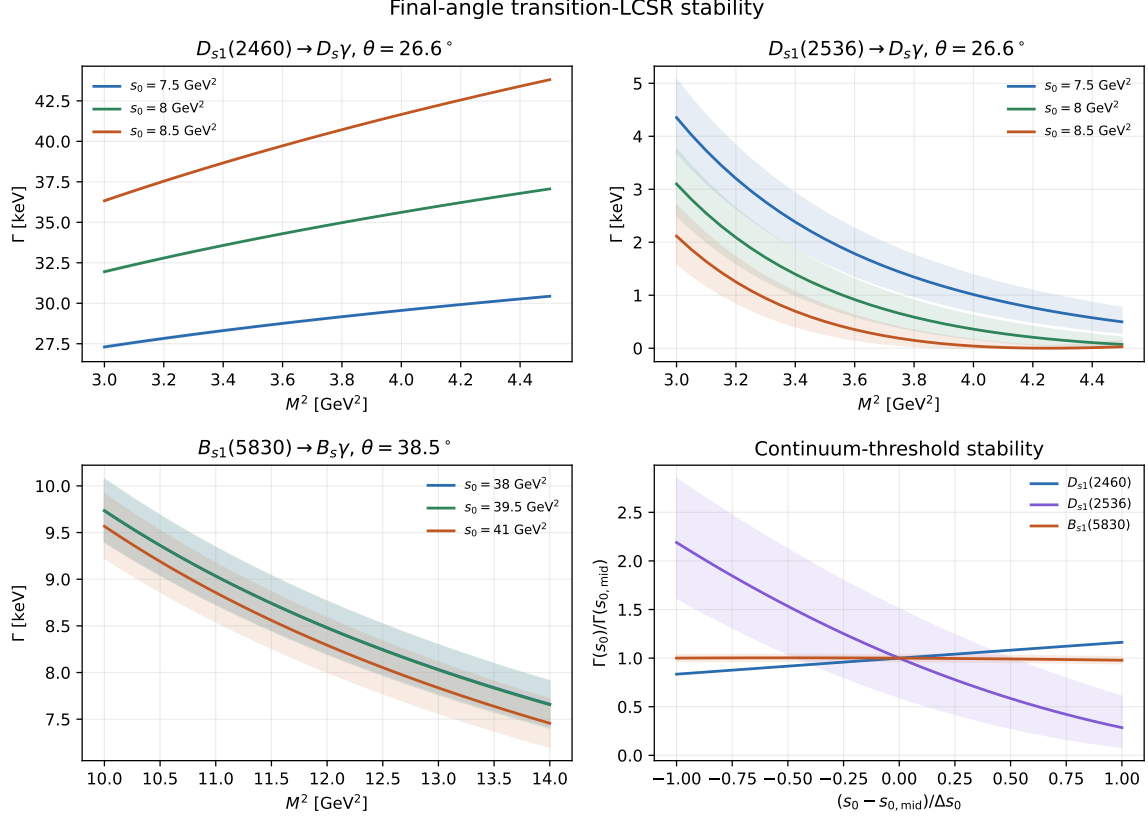


Figure 2: Transition-LCSR stability at the previous-study angles $\theta_{D_s} = 26.6 \pm 0.6^\circ$ and $\theta_{B_s} = 38.5 \pm 0.1^\circ$. The curves use the central angles and the shaded bands show only their quoted uncertainties. The physical residues are held at their accepted-sample medians to isolate Borel and threshold dependence. The high-state D_s channel is visibly more sensitive because its axial and tensor contributions nearly cancel.

Manuscript item	Source script	Source output	Purpose
input table	manual bookkeeping	inputs_table.csv	input values and citations
residue benchmark	stage3_residue_benchmark_checks.py	stage3_residue_benchmark_checks.csv	pure-current validation
main result table	Stage-3 charm and bottom scripts	combined_recommended_results_table.csv	final G and Γ values
final comparison table	paper summary scripts	bondar_literature_comparison_table.csv, experimental_comparison_table.csv	experiment and literature context
MC width plots	lattice/physical-normalization scan scripts	lattice_fperp...pdf, stage3_bs1_physical...pdf	uncertainty distributions
final-angle stability	final_angle_stability_plots.py	final_angle_stability.csv, final_angle_stability.pdf	Borel, threshold, and quoted-angle checks
validation checks	validation and Ward scripts	validation_checks_summary.txt, ward_transversality_check.txt	limiting cases and gauge structure
interference diagnostic	ds1_interference_diagnostic.py	ds1_interference_diagnostic_summary.csv	constructive/destructive amplitude pieces

Table 5: Reproducibility map for the headline material in this draft.

7 Conclusions

We have developed a staged light-cone QCD sum-rule framework for the radiative transitions of strange heavy axial mesons. The calculation includes hard photon emission from both quark lines, two-particle and three-particle photon distribution amplitudes, and explicit strange-quark-mass terms. The charm residue normalization has been separated from the basis-current OPE and checked by projection onto the pure axial channel. This projection is the meaningful comparison with Wang’s one-current calculation; reproducing Wang’s one-pole number is not used as a criterion for the correctness of the two mixed residues. The bottom residues are now obtained from the direct heavy-flavour continuation of the same normalized-current $AA/AB/BB$ matrix, rather than from basis-current decay constants or an overlap closure. The former QCD-side pole-mass numerator shortcut has been withdrawn. The complete off-shell double-Borel reduction given in the appendix has now been propagated through the numerical uncertainty scan. At the central charm and bottom checkpoints, 145 convolution, invariant, coupling, residue, and width quantities agree between the Python and Mathematica implementations within the stated 10^{-8} absolute-or-relative regression tolerance.

Table 6 gives the requested consolidated comparison with experiment and representative calculations. The $D_{s1}(2460)$ interval overlaps the upper edge of the original single-current LCSR range of Colangelo et al. [15]. Combining our preferred partial width with the PDG 2025-update branching fraction $\mathcal{B}(D_{s1}(2460) \rightarrow D_s \gamma) = 18 \pm 4\%$ gives an indicative total width near 0.19 MeV, comfortably below the direct 3.5 MeV upper limit [16]. For $D_{s1}(2536)$, the measured 0.92 ± 0.05 MeV total width implies $\mathcal{B}_\gamma \simeq 0.107\%$ at our median, with a 16–84 percentile range of approximately 0.010–0.386%. This is below the indirect $\mathcal{B}_\gamma < 0.9\%$, or $\Gamma_\gamma < 8$ keV, constraint inferred by Bondar [4]. The observed $B_{s1}(5830)$ has no direct radiative measurement, so that row remains a genuine prediction of the direct-normalization calculation.

Channel and this work [keV]	Experimental information	Selected theory [keV]	Assessment
$D_{s1}(2460) \rightarrow D_s \gamma$ 34.96[28.06, 43.53]	$\mathcal{B}_\gamma = 18(4)\%$; $\Gamma_{\text{tot}} < 3.5$ MeV [16]	LCSR 19–29 [15]; QM 3.53–17.2 [17, 18]	overlaps the upper edge of the earlier LCSR
$D_{s1}(2536) \rightarrow D_s \gamma$ 0.988[0.092, 3.555]	$\Gamma_{\text{tot}} = 0.92(5)$ MeV [16]; indirect $\Gamma_\gamma < 8$ [4]	1.6 ± 2.3 [19]; 7–61 (models, incl. 15 from Ref. [20]); 12 [5]	compatible with the bound; cancellation sensitive
$B_{s1}(5830) \rightarrow B_s \gamma$ 8.702[6.474, 11.23]	no direct radiative measurement [3]	27–56.6 [21, 22, 23]; 28 [5]	below existing models; cancellation sensitive

Table 6: Consolidated experiment and literature comparison for the preferred lattice-photon scenario. The wide theory spread reflects different state assignments, mixing conventions, and normalization schemes.

The scientifically robust conclusion is the interference pattern rather than an artificially precise high-state number. The $D_{s1}(2460)$ amplitude is constructive and naturally lies at the tens-of-keV level. The $D_{s1}(2536)$ amplitude is destructive and the final-angle stability plot shows enhanced threshold and angle sensitivity near its zero. The observed $B_{s1}(5830)$ is likewise a cancellation channel. Its direct two-point normalization raises the preferred width relative to the former closure diagnostic, but it remains below the representative quark-model predictions.

A Explicit analytic light-cone sum rules

A.1 Correlator, Wick contractions, and propagators

With $p' = p + q$, the Rohrwild-style external-photon correlator used for either basis current is

$$\Pi_\mu^X(p, q) = i \int d^4x e^{ip \cdot x} \langle \gamma(q, \varepsilon) | T \{ J_\mu^X(x) J_5^\dagger(0) \} | 0 \rangle, \quad X = A, B. \quad (16)$$

The hard-emission Wick contraction is

$$\Pi_\mu^{X, \text{hard}} = -i N_c \int \frac{d^4k}{(2\pi)^4} \text{Tr} \left[\Gamma_\mu^X S_Q^\gamma(k) i\gamma_5 S_s^0(k-p) + \Gamma_\mu^X S_Q^0(k) i\gamma_5 S_s^\gamma(k-p) \right], \quad (17)$$

where $S_f^0(k) = i(\not{k} + m_f)/(k^2 - m_f^2 + i0)$. The light-strange propagator is written without ordinary local-condensate pieces:

$$\begin{aligned} s^a(\widehat{x})\bar{s}^b(0) &= \delta^{ab} \left[\frac{i\not{x}}{2\pi^2 x^4} - \frac{m_s}{4\pi^2 x^2} \right] \\ &\quad - \frac{ig_s}{16\pi^2 x^2} \int_0^1 du \{ \bar{u}\not{x}\sigma_{\alpha\beta} + u\sigma_{\alpha\beta}\not{x} \} G_A^{\alpha\beta}(ux) t_{ab}^A \\ &\quad - \frac{ie_s \delta^{ab}}{16\pi^2 x^2} \int_0^1 du \{ \bar{u}\not{x}\sigma_{\alpha\beta} + u\sigma_{\alpha\beta}\not{x} \} F^{\alpha\beta}(ux) \\ &\quad - \frac{m_s}{32\pi^2} \int_0^1 du [g_s G_A^{\alpha\beta}(ux) t_{ab}^A + e_s \delta^{ab} F^{\alpha\beta}(ux)] \sigma_{\alpha\beta} L_x + \cdots, \end{aligned} \quad (18)$$

where $\bar{u} = 1 - u$ and $L_x = \ln(-x^2 \mu^2/4) + 2\gamma_E$. The last line is the linear- m_s Wilson coefficient in one background field; it is not a local condensate. Setting $m_s = 0$ reproduces the form used by Rohrwild [7].

The one-background-field momentum representation used for both heavy and light lines is

$$\begin{aligned} S_f^\mathcal{F}(x) &= -ic_f \int \frac{d^4k}{(2\pi)^4} e^{-ik \cdot x} \int_0^1 du \left[\frac{\not{k} + m_f}{2(m_f^2 - k^2)^2} \mathcal{F}^{\alpha\beta}(ux) \sigma_{\alpha\beta} \right. \\ &\quad \left. + \frac{ux_\alpha}{m_f^2 - k^2} \mathcal{F}^{\alpha\beta}(ux) \gamma_\beta \right], \end{aligned} \quad (19)$$

with $(c_f, \mathcal{F}) = (e_f, F)$ for the electromagnetic field and (g_s, G) for the gluon field. Equations (18)–(19) show explicitly that no ordinary local term such as $-\langle \bar{s}s \rangle/12$ is inserted into the transition calculation. The condensate-dependent terms instead occur in nonlocal photon matrix elements. In Rohrwild notation,

$$\begin{aligned} \langle 0 | \bar{s}(0) [0, x] \sigma_{\mu\nu} s(x) | 0 \rangle_F &= ee_s \langle \bar{s}s \rangle \int_0^1 du \left\{ \left[\chi_\gamma \phi_\gamma(u) + \frac{x^2}{16} \mathcal{A}(u) \right] F_{\mu\nu}(ux) \right. \\ &\quad \left. + \frac{\mathcal{B}(u)}{8} x^\rho [x_\nu F_{\mu\rho}(ux) - x_\mu F_{\nu\rho}(ux)] \right\}. \end{aligned} \quad (20)$$

The operator dictionary to the alternative notation is

$$\mathcal{B}(u) = -4 \int_0^u d\alpha (u - \alpha) h_\gamma(\alpha), \quad \mathbb{A}(u) - 8\bar{H}_\gamma(u) = \mathcal{A}(u) + 2\mathcal{B}(u). \quad (21)$$

In particular, the Rohrwild counterpart of the Colangelo $\bar{H}_\gamma$ term is $2\mathcal{B}$, not a new H_γ amplitude.

A.2 Double-Borel normalization and axial invariant

Define

$$\langle 0 | J_5 | P(p) \rangle = \frac{f_P m_P^2}{m_Q + m_s}, \quad \langle 0 | J_i^\mu | H_i(p', \eta) \rangle = f_i M_i \eta^\mu. \quad (22)$$

After projection onto $T_{\mu\nu} = (p \cdot q) g_{\mu\nu} - q_\mu p_\nu$, the pole contribution is

$$\hat{T}_i^{\text{pole}} = \frac{f_P m_P^2}{m_Q + m_s} f_i M_i G_i \exp\left(-\frac{M_i^2}{\mathcal{M}_1^2} - \frac{m_P^2}{\mathcal{M}_2^2}\right). \quad (23)$$

For $\mathcal{M}_1^2 = \mathcal{M}_2^2 = 2M^2$,

$$\begin{aligned} G_1 &= \mathcal{N}_1 (\sin \theta \hat{T}_A + \cos \theta \hat{T}_B), \\ G_2 &= \mathcal{N}_2 (\cos \theta \hat{T}_A - \sin \theta \hat{T}_B), \quad \mathcal{N}_i = \frac{e^{(M_i^2 + m_P^2)/(2M^2)} (m_Q + m_s)}{M_i f_i m_P^2 f_P}. \end{aligned} \quad (24)$$

Let $E_Q = e^{-m_Q^2/M^2}$, $E_0 = e^{-s_0/M^2}$, $\Delta E_Q = E_Q - E_0$, and $u_0 = 1/2$. In the positive-magnitude susceptibility convention used in the numerical code, the paper-default axial invariant is

$$\begin{aligned} \hat{T}_A^{\text{RW}} &= \int_{(m_Q + m_s)^2}^{s_0} ds e^{-s/M^2} \rho_A^P(s) + e_s \langle \bar{s}s \rangle \Delta E_Q M^2 \chi_\gamma \phi_\gamma(u_0) \\ &\quad + e_s \langle \bar{s}s \rangle E_Q \left[\frac{1}{4} \{ \mathcal{A}(u_0) + 2\mathcal{B}(u_0) \} \left(1 + \frac{m_Q^2}{M^2} \right) + \mathcal{I}_{\mathcal{F}_1}(u_0) \right] \\ &\quad + e_s f_{3\gamma} m_Q E_Q \bar{\psi}^{(V)}(u_0) + \hat{T}_A^{3p,\gamma}. \end{aligned} \quad (25)$$

Thus the continuum factors are not universal: twist 2 carries ΔE_Q , whereas twist 3, twist 4, and the gluonic three-particle term carry E_Q . The separate gauge-completion electromagnetic term below follows Rohrwild's I_F -type ΔE_Q subtraction. Here $\bar{\psi}^{(V)}(u) = 2 \int_0^u dv \psi^v(v)$, and

$$\begin{aligned} \mathcal{I}_{\mathcal{F}_1}(u_0) &= \int_0^{1-u_0} dv \int_0^{u_0/(1-v)} d\alpha_g \mathcal{F}_1(u_0 - (1-v)\alpha_g, 1 - u_0 - v\alpha_g, \alpha_g) \\ &\quad + \int_{1-u_0}^1 dv \int_0^{(1-u_0)/v} d\alpha_g \mathcal{F}_1(u_0 - (1-v)\alpha_g, 1 - u_0 - v\alpha_g, \alpha_g). \end{aligned} \quad (26)$$

The three-particle combination appearing here is

$$\mathcal{F}_1 = \mathcal{S} + \tilde{\mathcal{S}} - \mathcal{T}_1 - \mathcal{T}_2 + \mathcal{T}_3 + \mathcal{T}_4 + 2v(-\mathcal{S} - \mathcal{T}_3 + \mathcal{T}_2). \quad (27)$$

The perturbative density is unambiguously specified by

$$\begin{aligned} \rho_i(s; m_i, m_j, e_i) &= -\frac{3e_i}{8\pi^2} \left[2m_i \ln \frac{s - m_j^2 + m_i^2 - \sqrt{\lambda(s, m_j^2, m_i^2)}}{s - m_j^2 + m_i^2 + \sqrt{\lambda(s, m_j^2, m_i^2)}} \right. \\ &\quad \left. + (m_j - m_i) \frac{m_j^2 - m_i^2 - s}{s^2} \sqrt{\lambda(s, m_j^2, m_i^2)} \right], \\ \rho_A^P(s) &= \rho_s(s; m_s, m_Q, e_s) - \rho_Q(s; m_Q, m_s, e_Q). \end{aligned} \quad (28)$$

A.3 Tensor invariant and local-scheme separation

For completeness, the tensor hard density used in the diagonal double-dispersion prescription can be written

$$\begin{aligned}\rho_B^P(s) &= e_s \mathcal{L}(s; m_s, m_Q; a_s, b_s) - e_Q \mathcal{L}(s; m_Q, m_s; a_Q, b_Q), \\ \mathcal{L}(s; m_i, m_j; a, b) &= -\frac{3}{8\pi^2} \left[2a \ln \frac{s - m_j^2 + m_i^2 - \sqrt{\lambda}}{s - m_j^2 + m_i^2 + \sqrt{\lambda}} + \frac{b}{m_i^2} \frac{m_j^2 - m_i^2 - s}{s^2} \sqrt{\lambda} \right],\end{aligned}\quad (29)$$

where $d = m_Q + m_s$,

$$a_s = \frac{m_Q m_s}{d}, \quad b_s = \frac{m_s^2 s}{d}, \quad a_Q = \frac{2m_Q^2 - m_Q m_s}{d}, \quad b_Q = -\frac{m_Q^2 s}{d}. \quad (30)$$

On the QCD side, p'^2 and p^2 remain independent until the double Borel transformation:

$$q^2 = 0, \quad p' = p + q, \quad 2p \cdot q = p'^2 - p^2, \quad D(u) = m_Q^2 - \bar{u}p'^2 - up^2. \quad (31)$$

The substitutions $p'^2 = M_i^2$, $p^2 = m_P^2$ and $2p \cdot q = M_i^2 - m_P^2$ belong only to the isolated phenomenological pole residue and must not be made inside a QCD-side numerator.

The current traces give the exact kinematics-independent relations

$$\hat{T}_B^{\text{tw}2} = \frac{m_Q}{d} \hat{T}_A^{\text{tw}2}, \quad \hat{T}_B^{\text{tw}4} = \frac{m_Q}{d} \hat{T}_A^{\text{tw}4}. \quad (32)$$

The vector-DA and $\mathbb{P}$ three-particle terms contain p^2 or $p \cdot q$ and must therefore be transformed off shell. For $I_n[F] = \int_0^1 du F(u)/D(u)^n$, define

$$\frac{1}{M^2} = \frac{1}{\mathcal{M}_1^2} + \frac{1}{\mathcal{M}_2^2}, \quad u_0 = \frac{\mathcal{M}_1^2}{\mathcal{M}_1^2 + \mathcal{M}_2^2}, \quad A_n = \frac{(M^2)^{2-n}}{\Gamma(n)} e^{-m_Q^2/M^2}. \quad (33)$$

Then, for $n > 1$ and vanishing endpoint terms,

$$\begin{aligned}\mathcal{B}_{12} I_n[F] &= A_n F(u_0), \\ \mathcal{B}_{12} \int_0^1 du \frac{(p \cdot q) F(u)}{D(u)^n} &= \frac{A_{n-1}}{2(n-1)} F'(u_0), \\ \mathcal{B}_{12} \int_0^1 du \frac{p^2 F(u)}{D(u)^n} &= m_Q^2 A_n F(u_0) - A_{n-1} F(u_0) - \frac{A_{n-1}}{n-1} [\bar{u} F(u)]'_{u=u_0}.\end{aligned}\quad (34)$$

Nonvanishing endpoints generate additional contact terms. These identities replace the former pole-mass numerator prescription.

For the vector DA set $a = 1 - u$, $a_0 = 1 - u_0$ and $D_a = m_Q^2 - ap'^2 - (1-a)p^2$. The reduced pre-Borel kernels and the explicit tensor-current Borel result are

$$\begin{aligned}K_A^{(V)}(a) &= -\frac{8m_Q}{D_a^2}, \\ K_B^{(V)}(a) &= -\frac{8[m_Q^2 + (1-a)p \cdot q]}{dD_a^2}, \\ \hat{T}_B^{\text{tw}3} &= \frac{e_s f_{3\gamma} E_Q}{d} \left[m_Q^2 \bar{\psi}^{(V)}(a_0) - \frac{M^2}{2} \frac{d}{da} \{ (1-a) \bar{\psi}^{(V)}(a) \} \Big|_{a=a_0} \right].\end{aligned}\quad (35)$$

At $a_0 = 1/2$,

$$\hat{T}_A^{\text{tw}3} = 0, \quad \hat{T}_B^{\text{tw}3} = -\frac{e_s f_{3\gamma} M^2 E_Q}{4d} \bar{\psi}^{(V)'}(1/2). \quad (36)$$

For the three-particle pieces the heavy-line fraction is $a(\alpha_i, v) = \alpha_{\bar{q}} + v\alpha_g$. After performing the delta-function integration over the physical simplex, define

$$\begin{aligned}\mathcal{J}_z[\mathcal{F}] &= \int_0^z dv \int_0^{(1-z)/(1-v)} d\alpha_g \mathcal{F}(1-z-(1-v)\alpha_g, z-v\alpha_g, \alpha_g, v) \\ &\quad + \int_z^1 dv \int_0^{z/v} d\alpha_g \mathcal{F}(1-z-(1-v)\alpha_g, z-v\alpha_g, \alpha_g, v).\end{aligned}\quad (37)$$

The $1/(q \cdot x)$ terms require the support density

$$\begin{aligned}\mathcal{L}_z[X] &= \int_0^{1-z} d\alpha_q \int_0^{\alpha_q} d\alpha'_q \int_0^z d\alpha_{\bar{q}} \frac{z - \alpha_{\bar{q}}}{(1 - \alpha_q - \alpha_{\bar{q}})^2} X(\alpha_q, \alpha_{\bar{q}}, 1 - \alpha_q - \alpha_{\bar{q}}) \\ &\quad - \int_0^{1-z} d\alpha_q \int_z^{1-\alpha_q} d\beta \frac{z}{\beta^2} \int_0^\beta d\alpha'_{\bar{q}} X(\alpha_q, \alpha'_{\bar{q}}, 1 - \alpha_q - \alpha'_{\bar{q}}) \\ &\quad + \int_0^{1-z} d\alpha_q \frac{z}{(1 - \alpha_q)^2} \int_0^{\alpha_q} d\alpha'_q \int_0^{1-\alpha'_q} d\alpha_{\bar{q}} X(\alpha'_q, \alpha_{\bar{q}}, 1 - \alpha'_q - \alpha_{\bar{q}}).\end{aligned}\quad (38)$$

The lower limit $\beta = z$ in the second line is required by the double-Borel support; extending it to zero produces a spurious logarithmic divergence.

The $x_\alpha G^{\alpha\beta} \gamma_\beta$ heavy-propagator term has no $(\not{k} + m_Q)$ numerator. Its explicit E1 trace therefore gives

$$\begin{aligned}\mathfrak{X}_{A,S}[F] &= -8E_Q F(a_0), & \mathfrak{X}_{A,T_1}[F] &= 48iE_Q F(a_0), \\ \mathfrak{X}_{A,T_2}[F] &= 16iE_Q F(a_0), & \mathfrak{X}_{A,T_3}[F] &= -16iE_Q F(a_0), \\ \mathfrak{X}_{A,T_4}[F] &= 0, & & \\ \mathfrak{X}_{B,S}[F] &= \mathfrak{X}_{B,T_1}[F] = \mathfrak{X}_{B,T_2}[F] = \mathfrak{X}_{B,T_3}[F] = \mathfrak{X}_{B,T_4}[F] = 0.\end{aligned}\quad (39)$$

The corresponding $\sigma_{\alpha\beta}$ coefficients are $\{8, -16i\}$ for A and $\{8m_Q/d, -16im_Q/d\}$ for B . Equations (39)–(40) are raw Borel images. The final continuum prescription is term-specific: twist 2 uses ΔE_Q , twist 3, twist 4, and the gluonic three-particle term retain E_Q , and the separate gauge-completion electromagnetic term uses ΔE_Q .

Suppressing the arguments $(\alpha_q, \alpha_{\bar{q}}, \alpha_g)$, the complete gluonic operator weights are

$$\begin{aligned}\mathcal{F}_{A,\sigma}^g &= \mathcal{S} + \tilde{\mathcal{S}} - \mathcal{T}_1 - \mathcal{T}_2 + \mathcal{T}_3 + \mathcal{T}_4, \\ \mathcal{F}_{A,x\gamma}^g &= 2v(-\mathcal{S} - \mathcal{T}_3 + \mathcal{T}_2), \\ \mathcal{F}_A^g &= \mathcal{F}_{A,\sigma}^g + \mathcal{F}_{A,x\gamma}^g, & \mathcal{F}_B^g &= \frac{m_Q}{d} \mathcal{F}_{A,\sigma}^g.\end{aligned}\quad (41)$$

Their explicit post-Borel invariants are

$$\hat{T}_A^{3p,g} = e_s \langle \bar{s}s \rangle E_Q \mathcal{J}_{a_0}[\mathcal{F}_A^g], \quad \hat{T}_B^{3p,g} = e_s \langle \bar{s}s \rangle E_Q \mathcal{J}_{a_0}[\mathcal{F}_B^g]. \quad (42)$$

For the nonlocal electromagnetic operators,

$$\mathcal{F}_A^\gamma = (1 - 2v)\mathcal{S}_\gamma - \mathcal{T}_4^\gamma, \quad \mathcal{F}_B^\gamma = \frac{m_Q}{d}(\mathcal{S}_\gamma - \mathcal{T}_4^\gamma). \quad (43)$$

The exact off-shell Borel transform replaces the former pole-mass numerator by

$$\frac{2(m_P^2 - M_i^2)}{M^2} \mathcal{L}_{a_0}[\mathcal{T}_4^\gamma] \longrightarrow 2 \frac{d}{dz} \mathcal{L}_z[\mathcal{T}_4^\gamma] \Big|_{z=a_0}. \quad (44)$$

Writing $\Delta E_Q = E_Q - E_0$, the complete electromagnetic terms are

$$\begin{aligned}\hat{T}_A^{3p,\gamma} &= e_Q \langle \bar{s}s \rangle \Delta E_Q \left\{ \mathcal{J}_{a_0}[(1 - 2v)\mathcal{S}_\gamma - \mathcal{T}_4^\gamma] + 2 \frac{d}{dz} \mathcal{L}_z[\mathcal{T}_4^\gamma] \Big|_{z=a_0} \right\}, \\ \hat{T}_B^{3p,\gamma} &= e_Q \langle \bar{s}s \rangle \Delta E_Q \frac{m_Q}{d} \left\{ \mathcal{J}_{a_0}[\mathcal{S}_\gamma - \mathcal{T}_4^\gamma] + 2 \frac{d}{dz} \mathcal{L}_z[\mathcal{T}_4^\gamma] \Big|_{z=a_0} \right\}.\end{aligned}\quad (45)$$

Thus the Rohrwild tensor invariant has the parallel structure

$$\hat{T}_B^{\text{RW}} = \int ds e^{-s/M^2} \rho_B^P + \hat{T}_B^{\text{tw}2} + \hat{T}_B^{\text{tw}3} + \hat{T}_B^{\text{tw}4} + \hat{T}_B^{3\text{p},g} + \hat{T}_B^{3\text{p},\gamma}. \quad (46)$$

Equations (37)–(45) display all integration limits, operator weights and Borel derivatives. No physical pole mass and no ordinary local condensate enters a QCD-side numerator.

A.4 Numerical convolution and cross-code checkpoint

At the symmetric saddle the DA-only convolutions are

$$\begin{aligned} \mathcal{J}_{1/2}[\mathcal{F}_{A,\sigma}^g] &= -0.1562500000, & \mathcal{J}_{1/2}[\mathcal{F}_{A,x\gamma}^g] &= +0.2265798500, \\ \mathcal{J}_{1/2}[\mathcal{F}_A^g] &= +0.0703298500, & \mathcal{J}_{1/2}[\mathcal{F}_A^\gamma] &= 0, \\ \mathcal{J}_{1/2}[\mathcal{S}_\gamma - \mathcal{T}_4^\gamma] &= 1.5625000000, & \mathcal{L}_{1/2}[\mathcal{T}_4^\gamma] &= 0.1111834583, \\ \left. \frac{d}{dz} \mathcal{L}_z[\mathcal{T}_4^\gamma] \right|_{z=1/2} &= 0.4098669166. \end{aligned} \quad (47)$$

For the central charm regression point $M^2 = 3.75 \text{ GeV}^2$, $s_0 = 8.0 \text{ GeV}^2$, the lattice photon normalization, $\theta = 26.6^\circ$, and $(f_1, f_2) = (0.4005893, 0.1666704) \text{ GeV}$, the invariant breakdown is

contribution	$\hat{T}_A \text{ (GeV}^3\text{)}$	$\hat{T}_B \text{ (GeV}^3\text{)}$
perturbative	$-8.8661701 \times 10^{-2}$	$-1.1226165 \times 10^{-1}$
twist 2	$+5.4942081 \times 10^{-2}$	$+5.1193282 \times 10^{-2}$
twist 3	0	$-5.2253296 \times 10^{-3}$
twist 4	$+5.5991083 \times 10^{-3}$	$+5.2170708 \times 10^{-3}$
three-particle gluonic	$+1.6863557 \times 10^{-4}$	$-3.4908999 \times 10^{-4}$
three-particle electromagnetic	$-3.2152535 \times 10^{-3}$	$-8.7063197 \times 10^{-3}$
total	$-3.1167129 \times 10^{-2}$	$-7.0132040 \times 10^{-2}$

Table 7: Central invariant checkpoint. The corresponding single-point couplings are $G_1 = -0.4113424 \text{ GeV}^{-1}$ and $G_2 = +0.0464987 \text{ GeV}^{-1}$, giving 35.5665 and 0.67094 keV. These are regression values, not the Monte Carlo medians in Table 3.

The independent Python and Mathematica tables contain 145 common numerical entries and pass an absolute-or-relative 10^{-8} comparison. The executable sources and machine-readable residuals are, respectively,

scripts/corrected_transition_central_python.py,
scripts/mathematica_corrected_transition_numerics.wl,
outputs/corrected_transition_python_mathematica_comparison.csv.

Contribution	Rohrwild default	Colangelo comparison
ordinary local transition condensate	absent	retained for J_A
$\mathcal{S}_\gamma, \mathcal{T}_4^\gamma$	included nonlocally	not added separately
gluonic photon DAs	included	included through $\mathcal{F}_1$
local condensates in two-point f_i sum rules	required	required
local plus electromagnetic term together	forbidden double counting	forbidden double counting

Table 8: Separation of the transition and two-point condensate prescriptions.

The alternative Colangelo axial invariant is obtained by removing $\hat{T}_A^{3p,\gamma}$ from Eq. (25) and adding

$$e_Q E_Q \langle \bar{s}s \rangle \left[1 - \frac{m_Q m_s}{M^2} + \frac{m_s^2}{2M^2} \left(1 + \frac{m_Q^2}{M^2} \right) \right]. \quad (48)$$

The two terms are alternatives and are never summed. In our rotation convention the exact pure-axial Colangelo limit is

$$\theta = \frac{\pi}{2}, \quad J_1 = J_A, \quad G_1 = \frac{e^{(M_1^2 + m_P^2)/(2M^2)} (m_Q + m_s)}{M_1 f_A m_P^2 f_P} \hat{T}_A. \quad (49)$$

Thus the agreement in analytic structure is exact at the pure-current point; with the Colangelo inputs our numerical pure- A width interval is 20.51–33.26 keV, overlapping their 19–29 keV result.

B Explicit $AA/AB/BB$ two-point sum rules

For $i, j \in \{A, B\}$,

$$\Pi_{ij}^{\mu\nu}(p') = i \int d^4x e^{ip' \cdot x} \langle 0 | T \{ J_i^\mu(x) J_j^{\nu\dagger}(0) \} | 0 \rangle = \left(-g^{\mu\nu} + \frac{p'^\mu p'^\nu}{p'^2} \right) \Pi_{ij}(p'^2) + \dots \quad (50)$$

Let $d = m_Q + m_s$ and $\lambda(s) = [s - (m_Q + m_s)^2][s - (m_Q - m_s)^2]$. The exact-mass LO spectral matrix is

$$\begin{aligned} \rho_{AA}(s) &= \frac{\sqrt{\lambda}}{8\pi^2} \left[2 - \frac{m_Q^2 + m_s^2 + 6m_Q m_s}{s} - \frac{(m_Q^2 - m_s^2)^2}{s^2} \right], \\ \rho_{AB}(s) &= \rho_{BA}(s) = \frac{\sqrt{\lambda}}{8\pi^2} \frac{3(m_s - m_Q)(d^2 - s)}{ds}, \\ \rho_{BB}(s) &= \frac{\sqrt{\lambda}}{8\pi^2} \frac{-(d^2 - s)(2m_s^2 - 4m_Q m_s + 2m_Q^2 + s)}{d^2 s}. \end{aligned} \quad (51)$$

With $C_3 = \langle \bar{s}s \rangle$ and $C_5 = \langle \bar{s}g_s \sigma \cdot G s \rangle$, the common LO+ $d = 3+d = 5$ Borel matrix is

$$\hat{\Pi}(M^2, s_0) = \int_{d^2}^{s_0} ds e^{-s/M^2} \rho(s) + e^{-m_Q^2/M^2} [C_3(\mathcal{Q}_0 + \mathcal{Q}_1 + \mathcal{Q}_2) + C_5 \mathcal{Q}_5], \quad (52)$$

where

$$\begin{aligned} \mathcal{Q}_0 &= \begin{pmatrix} m_Q & m_Q^2/d \\ m_Q^2/d & m_Q^3/d^2 \end{pmatrix}, \\ \mathcal{Q}_1 &= \begin{pmatrix} -\frac{m_s m_Q^2}{2M^2} & \frac{m_s m_Q (1 - m_Q^2/M^2)}{2d} \\ \frac{m_s m_Q (1 - m_Q^2/M^2)}{2d} & \frac{m_s m_Q^2 (1 - m_Q^2/(2M^2))}{d^2} \end{pmatrix}, \\ \mathcal{Q}_2 &= \begin{pmatrix} \frac{m_s^2 m_Q^3}{2M^4} & \frac{m_s^2 (m_Q^4/M^4 - m_Q^2/M^2 - 1)}{2d} \\ \frac{m_s^2 (m_Q^4/M^4 - m_Q^2/M^2 - 1)}{2d} & \frac{m_s^2 (m_Q^5/(2M^4) - m_Q^3/M^2)}{d^2} \end{pmatrix}, \\ \mathcal{Q}_5 &= \begin{pmatrix} -\frac{m_Q^3}{4M^4} & -\frac{m_Q^4/M^4 - m_Q^2/M^2 - 1}{4d} \\ -\frac{m_Q^4/M^4 - m_Q^2/M^2 - 1}{4d} & -\frac{m_Q^5/(4M^4) - m_Q^3/(2M^2)}{d^2} \end{pmatrix}. \end{aligned} \quad (53)$$

These local condensates are required because this is an ordinary two-point SVZ sum rule; their use does not contradict their exclusion from the external-photon transition LCSR.

Writing $s_\theta = \sin \theta$ and $c_\theta = \cos \theta$, the physical invariants are

$$\begin{aligned}\hat{\Pi}_{11} &= s_\theta^2 \hat{\Pi}_{AA} + c_\theta^2 \hat{\Pi}_{BB} + 2s_\theta c_\theta \hat{\Pi}_{AB}, \\ \hat{\Pi}_{22} &= c_\theta^2 \hat{\Pi}_{AA} + s_\theta^2 \hat{\Pi}_{BB} - 2s_\theta c_\theta \hat{\Pi}_{AB}, \\ \hat{\Pi}_{12} &= s_\theta c_\theta (\hat{\Pi}_{AA} - \hat{\Pi}_{BB}) + (c_\theta^2 - s_\theta^2) \hat{\Pi}_{AB}.\end{aligned}\tag{54}$$

For $n = 0, 1, 2, 5$, define the fully specified scalar projections of the matrices in Eq. (53) by

$$\begin{aligned}q_1^{(n)} &= s_\theta^2 (\mathcal{Q}_n)_{AA} + c_\theta^2 (\mathcal{Q}_n)_{BB} + 2s_\theta c_\theta (\mathcal{Q}_n)_{AB}, \\ q_2^{(n)} &= c_\theta^2 (\mathcal{Q}_n)_{AA} + s_\theta^2 (\mathcal{Q}_n)_{BB} - 2s_\theta c_\theta (\mathcal{Q}_n)_{AB}.\end{aligned}\tag{55}$$

The two pole channels use separate thresholds. The expressions actually evaluated are

$$\begin{aligned}\hat{\Pi}_{11}^{(1)} &= \int_{d^2}^{s_0^{(1)}} ds e^{-s/M^2} [s_\theta^2 \rho_{AA} + c_\theta^2 \rho_{BB} + 2s_\theta c_\theta \rho_{AB}] \\ &\quad + e^{-m_Q^2/M^2} \{C_3[q_1^{(0)} + q_1^{(1)} + q_1^{(2)}] + C_5 q_1^{(5)}\}, \\ \hat{\Pi}_{22}^{(2)} &= \int_{d^2}^{s_0^{(2)}} ds e^{-s/M^2} [c_\theta^2 \rho_{AA} + s_\theta^2 \rho_{BB} - 2s_\theta c_\theta \rho_{AB}] \\ &\quad + e^{-m_Q^2/M^2} \{C_3[q_2^{(0)} + q_2^{(1)} + q_2^{(2)}] + C_5 q_2^{(5)}\}.\end{aligned}\tag{56}$$

Consequently,

$$\begin{aligned}\boxed{f_1(M^2, s_0^{(1)}, \theta)} &= \frac{e^{M_1^2/(2M^2)}}{M_1} \sqrt{\hat{\Pi}_{11}^{(1)}}, \\ \boxed{f_2(M^2, s_0^{(2)}, \theta)} &= \frac{e^{M_2^2/(2M^2)}}{M_2} \sqrt{\hat{\Pi}_{22}^{(2)}}.\end{aligned}\tag{57}$$

We insert the previous-study angle $26.6 \pm 0.6^\circ$ [9] here; we do not refit it in the radiative paper. Setting $\hat{\Pi}_{12} = 0$ gives the $30.63[29.94, 31.29]^\circ$ truncation diagnostic quoted in the text. At the central point $M^2 = 2.35 \text{ GeV}^2$, $s_0^{(1)} = 7.925 \text{ GeV}^2$ and $s_0^{(2)} = 9.975 \text{ GeV}^2$, these expressions give $f_1 = 0.4005893 \text{ GeV}$ and $f_2 = 0.1666704 \text{ GeV}$. The values quoted in the results tables are the corresponding Monte Carlo medians and 16–84 percentile intervals.

For the bottom channel the same formulas are evaluated with $m_Q = m_b$, $\theta_{B_s} = 38.5^\circ$, $M^2 = 7.00 \text{ GeV}^2$, $s_0^{(1)} = 44.35 \text{ GeV}^2$, and $s_0^{(2)} = 43.025 \text{ GeV}^2$. The central direct residues are

$$f_1^{(B_s)} = 0.5381107 \text{ GeV}, \quad f_2^{(B_s)} = 0.0888787 \text{ GeV}.\tag{58}$$

The corresponding pole fractions are 0.805 and 0.556, while the dimension-five fractions are 0.0029 and 0.0348. Diagonalizing the truncated bottom matrix gives 39.50° as a diagnostic, close to the external angle; the normalized off-diagonal residual is 0.104. Varying the OPE inputs and the two-point Borel parameter produces the Monte Carlo intervals quoted in the text.

B.1 Projection onto Wang’s pure axial channel

Inverting the physical rotation gives $J_A = s_\theta J_1 + c_\theta J_2$. Consequently the two-pole hadronic representation of the pure- AA invariant is

$$\hat{\Pi}_{AA}^{(2\text{pole})} = s_\theta^2 f_1^2 M_1^2 e^{-M_1^2/M^2} + c_\theta^2 f_2^2 M_2^2 e^{-M_2^2/M^2}.\tag{59}$$

For comparison with a one-pole result normalized at M_1 , define only as a diagnostic

$$f_{A,\text{eq}}^2(M^2) = \frac{e^{M_1^2/M^2}}{M_1^2} \hat{\Pi}_{AA}^{(2\text{ pole})}. \quad (60)$$

The accepted two-point samples give $f_{A,\text{eq}} = 0.2303[0.2194, 0.2418]$ GeV, while the separate direct AA OPE in Wang's window gives $0.2686[0.2607, 0.2768]$ GeV and Wang quotes 0.345 ± 0.017 GeV [10]. The three quantities test related normalization scales but are not identical calculations: the first uses two physical poles, the second uses our stated truncated OPE, and the third uses Wang's one-pole OPE. This is why the comparison is useful but cannot by itself prove or disprove the mixed-current result.

Acknowledgments

To be added.

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